

BANGALURU INTL AP, IndiaWMO: **427056**Lat: **13.200N**Lon: **77.700E**Elev: **3002**StdP: **13.17**Time Zone: **5.50 (IND)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.4	55.5	30.3	27.2	84.8	35.4	33.8	83.8	15.3	78.1	14.1	77.9	1.5	100	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.4	98.9	69.0	96.5	69.2	93.5	68.4	79.8	84.0	78.8	83.1	77.8	82.8	7.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	168.6	81.1	77.4	159.8	80.2	77.0	157.4	79.9	46.1	84.9	45.2	81.6	44.1	83.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.7	19.1	17.0	49.6	97.9	3.0	3.8	47.4	100.7	45.6	102.9	43.9	105.1	41.8	107.9	
(5)				48.4	78.4	3.0	3.6	46.2	81.0	44.4	83.1	42.7	85.1	40.5	87.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	70.6	73.9	79.5	83.3	82.3	78.7	77.3	76.6	76.7	75.7	73.2	71.2
	DBStd	5.24	3.85	4.24	4.21	3.95	3.33	3.31	3.13	3.14	3.00	3.30	3.40	3.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD50	9706	638	668	913	1000	1002	860	846	824	802	798	696	659
	CDD65	4232	174	248	448	550	537	410	381	359	352	333	246	194
	CDH74	38075	1615	2868	5501	6774	5814	3373	2595	2246	2428	2243	1409	1209
	CDH80	15423	405	1220	2883	3761	2830	1171	737	580	672	621	300	243
Wind	WSAvg	7.9	5.0	5.6	6.0	6.0	8.0	12.9	13.8	11.5	9.1	5.7	5.4	5.6
Precipitation	PrecAvg	37.9	0.2	0.2	0.5	1.3	3.2	3.5	4.5	5.6	5.7	7.4	4.4	1.6
	PrecMax	55.6	1.9	2.4	4.1	4.1	9.8	10.4	8.3	11.2	9.0	19.9	17.1	6.6
	PrecMin	26.7	0.0	0.0	0.0	0.1	0.4	1.2	1.3	1.8	2.0	0.8	0.2	0.0
	PrecStd	7.8	0.4	0.5	0.8	0.9	2.0	2.0	1.9	2.2	1.9	3.5	3.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	95.3	100.3	103.6	99.0	93.6	91.8	91.5	91.6	91.6	88.0	88.8
		MCWB	67.8	66.0	68.5	70.6	72.2	75.5	74.9	76.4	76.1	72.6	71.2	69.9
	2%	DB	86.3	92.8	97.2	100.5	96.6	91.6	89.5	88.2	89.3	89.2	85.8	85.8
		MCWB	65.5	64.8	66.3	69.4	73.0	75.3	75.8	75.9	76.0	74.5	69.5	69.3
	5%	DB	84.3	89.9	94.8	97.0	93.5	89.2	86.3	85.8	86.1	85.7	83.8	82.5
		MCWB	64.0	63.6	65.7	70.0	71.6	74.8	73.2	74.1	73.8	72.0	68.8	66.9
	10%	DB	82.2	87.5	91.6	93.6	91.5	86.1	84.1	82.7	83.9	82.8	81.0	80.4
		MCWB	62.5	62.0	64.4	66.5	71.4	72.8	71.8	72.0	72.5	69.9	67.6	65.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	75.6	78.9	80.9	80.9	79.7	78.9	79.2	80.6	79.8	79.1	78.4
		MCDB	77.4	79.8	82.4	84.7	86.7	84.1	83.2	82.4	84.1	82.8	81.4	80.1
	2%	WB	73.4	73.0	76.0	78.3	79.4	78.4	77.4	78.0	78.7	78.5	77.2	75.4
		MCDB	77.6	77.5	82.2	85.0	85.0	83.5	82.4	82.5	83.0	82.3	79.6	78.7
	5%	WB	71.3	70.5	74.3	75.3	77.7	77.3	76.4	76.9	77.2	77.1	75.2	73.6
		MCDB	76.8	77.7	80.8	84.9	83.9	83.4	81.9	82.5	82.7	80.9	77.5	77.3
	10%	WB	68.5	68.3	72.1	73.4	76.4	76.3	75.4	75.6	75.6	75.4	72.9	71.4
		MCDB	75.7	76.9	79.9	83.5	83.4	82.9	81.1	80.8	80.1	78.7	75.7	75.0
Mean Daily Temperature Range	5% DB	MDBR	24.1	27.1	25.3	22.4	18.9	14.7	13.5	13.0	14.1	15.6	16.2	18.8
		MCDBR	24.9	28.5	26.7	23.8	20.6	16.8	15.6	15.2	16.1	17.7	19.1	21.2
		MCWBR	7.9	8.3	9.0	10.4	6.8	4.6	4.1	4.1	4.9	6.0	6.5	7.3
	5% WB	MCDBR	20.1	23.3	21.5	22.2	18.4	15.4	14.3	14.2	14.8	15.2	13.2	15.9
		MCWBR	7.6	8.7	9.1	10.6	7.2	5.3	4.3	4.4	5.8	6.1	6.6	6.9
Clear-Sky Solar Irradiance	taub	0.382	0.403	0.466	0.509	0.526	0.458	0.469	0.457	0.444	0.474	0.412	0.390	
	taud	2.306	2.247	2.076	2.001	1.984	2.230	2.195	2.237	2.269	2.160	2.314	2.327	
	Ebn at Noon	290	288	272	258	250	264	262	268	274	264	279	284	
	Edh at Noon	40	44	54	58	58	45	47	45	44	48	40	39	
All-Sky Solar Radiation	RadAvg	1705	1953	2071	2091	1995	1679	1496	1513	1631	1573	1465	1496	
	RadStd	76	81	72	69	92	121	92	88	90	118	132	107	

Historical Trends

Station Only	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.58	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+308	+310